



Synthetic Analytics & Alerting

Series: SYNTH | **Notebook:** 6 of 6 | **Created:** December 2025

Dashboards, SLOs, and Alerting Strategies

This notebook covers advanced analytics for synthetic monitoring, including building dashboards, configuring SLOs, and implementing effective alerting strategies using the latest Dynatrace platform.

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Prerequisites

-  Access to a Dynatrace environment with Synthetic Monitoring
-  Completed SYNTH-01 through SYNTH-05
-  Active synthetic monitors generating data

1. Analytics Overview

Key Metrics for Synthetic Monitoring

Metric Category	Metrics	Use Case
Availability	Success rate, failure count	SLA compliance
Performance	Response time, TTFB, DNS	User experience
Reliability	Consecutive failures, MTTR	Stability
Geographic	Per-location metrics	Regional issues

Data Sources

Source	Description	Best For
bizevents	Execution results	Real-time analysis
dt.entity.synthetic_test	Monitor definitions	Configuration
dt.entity.synthetic_location	Location info	Geographic analysis
metrics	Aggregated metrics	Long-term trends

2. Availability Analysis

Calculating Availability

Availability = (Successful Executions / Total Executions) × 100%

Example SLA Targets:

- 99.9% = max 8.76 hours downtime/year

- 99.5% = max 43.8 hours downtime/year

- 99.0% = max 87.6 hours downtime/year

```
// Overall synthetic availability (last 7 days)
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total_executions = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
}
| fieldsAdd availability_pct = round((successful * 100.0) / total_executions,
decimals: 3)
| fieldsAdd downtime_minutes = round((failed * 5.0), decimals: 0) //
Assuming 5-min frequency
```

```
// Availability by monitor (last 7 days)
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
}, by: {dt.entity.synthetic_test}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals:
3)
| sort availability_pct asc
| limit 30
```

```
// Availability trend over time
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd hour_bucket = bin(timestamp, 1h)
| summarize {
    success_count = countIf(synthetic.availability == true),
    total_count = count()
}, by: {hour_bucket}
| fieldsAdd availability_pct = round((success_count * 100.0) / total_count,
decimals: 2)
| sort hour_bucket desc
```

```
// Daily availability report
fetch bizevents, from: now() - 30d
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd day = formatTimestamp(timestamp, format: "yyyy-MM-dd")
| summarize {
    total = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
}, by: {day}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals:
3)
| sort day desc
| limit 30
```

3. Performance Analysis

Performance Metrics

Metric	Description	Good	Warning	Critical
Response Time	Total execution time	< 2s	2-5s	> 5s
DNS	DNS resolution	< 50ms	50-200ms	> 200ms
Connect	TCP connection	< 100ms	100-300ms	> 300ms
TTFB	Time to first byte	< 500ms	500ms-1s	> 1s

```
// Response time percentiles by monitor
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    p50_ms = percentile(toDouble(synthetic.response_time), 50),
    p75_ms = percentile(toDouble(synthetic.response_time), 75),
    p90_ms = percentile(toDouble(synthetic.response_time), 90),
    p95_ms = percentile(toDouble(synthetic.response_time), 95),
    p99_ms = percentile(toDouble(synthetic.response_time), 99),
    executions = count()
}, by: {dt.entity.synthetic_test}
| sort p95_ms desc
| limit 20
```

```
// Performance trend comparison (this week vs last week)
fetch bizevents, from: now() - 14d
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| fieldsAdd week = if(timestamp > now() - 7d, "This Week", else: "Last Week")
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    p95_response_ms = percentile(toDouble(synthetic.response_time), 95),
    executions = count()
}, by: {dt.entity.synthetic_test, week}
| sort dt.entity.synthetic_test, week
```

```
// Timing breakdown analysis
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    avg_dns_ms = avg(toDouble(synthetic.dns_time)),
    avg_connect_ms = avg(toDouble(synthetic.connect_time)),
    avg_ssl_ms = avg(toDouble(synthetic.ssl_time)),
    avg_ttfb_ms = avg(toDouble(synthetic.time_to_first_byte)),
    avg_download_ms = avg(toDouble(synthetic.download_time)),
    avg_total_ms = avg(toDouble(synthetic.response_time))
}, by: {dt.entity.synthetic_test}
| fieldsAdd other_ms = avg_total_ms - avg_dns_ms - avg_connect_ms -
avg_ssl_ms - avg_ttfb_ms - avg_download_ms
| sort avg_total_ms desc
| limit 20
```

```
// Performance anomalies (response time > 2x average)
// First calculate average per monitor, then find executions above threshold
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| fieldsAdd response_ms = toDouble(synthetic.response_time)
| summarize {
    avg_response = avg(response_ms),
    max_response = max(response_ms),
    min_response = min(response_ms),
    executions = count()
}, by: {dt.entity.synthetic_test}
| fieldsAdd deviation_factor = round(max_response / avg_response, decimals:
1)
| filter deviation_factor > 2
| sort deviation_factor desc
| limit 50
```

4. Location Comparison

Geographic Analysis

Compare synthetic results across locations to:

- Identify regional performance issues
- Detect CDN or DNS problems
- Validate global availability
- Optimize user experience by region

```
// Availability by location
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
}, by: {dt.entity.synthetic_location}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals:
2)
| sort availability_pct asc
| limit 30
```

```
// Performance by location
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    p50_ms = percentile(toDouble(synthetic.response_time), 50),
    p95_ms = percentile(toDouble(synthetic.response_time), 95),
    executions = count()
}, by: {dt.entity.synthetic_location}
| sort avg_response_ms desc
| limit 30
```

```
// Location performance heatmap (monitor x location)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    availability_pct = countIf(synthetic.availability == true) * 100.0 /
count()
}, by: {dt.entity.synthetic_test, dt.entity.synthetic_location}
| sort dt.entity.synthetic_test, avg_response_ms desc
```

```
// Location-specific failures
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| summarize {
    failure_count = count(),
    unique_monitors = countDistinct(dt.entity.synthetic_test),
    error_types = collectDistinct(synthetic.error_message)
}, by: {dt.entity.synthetic_location}
| sort failure_count desc
| limit 20
```

5. SLO Configuration

Service Level Objectives for Synthetic

SLO Type	Metric	Example Target
Availability	Success rate	99.9%
Performance	Response time P95	< 3000ms
Error Budget	Allowed failures	0.1% of executions

Creating Synthetic SLOs

1. **Navigate to:** Settings → Service-level objectives
2. **Create SLO:** Name, description, timeframe
3. **Define metric:** Use synthetic availability or response time
4. **Set target:** e.g., 99.9% availability
5. **Configure alerting:** Error budget alerts

```
// SLO calculation - Availability (30-day window)
fetch bizevents, from: now() - 30d
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total_executions = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
}, by: {dt.entity.synthetic_test}
| fieldsAdd sli_availability = round((successful * 100.0) / total_executions,
decimals: 4)
| fieldsAdd slo_target = 99.9
| fieldsAdd error_budget_total = round(total_executions * 0.001, decimals: 0)
// 0.1% budget
| fieldsAdd error_budget_remaining = round(error_budget_total - failed,
decimals: 0)
| fieldsAdd error_budget_pct = round((error_budget_remaining * 100.0) /
error_budget_total, decimals: 1)
| fieldsAdd slo_status = if(sli_availability >= slo_target, "✅ Met",
else: "❌ Breached")
| sort sli_availability asc
| limit 30
```



```
// SLO calculation - Performance (P95 < 3000ms)
fetch bizevents, from: now() - 30d
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    executions = count(),
    p95_response_ms = percentile(toDouble(synthetic.response_time), 95),
    within_target = countIf(toDouble(synthetic.response_time) < 3000)
}, by: {dt.entity.synthetic_test}
| fieldsAdd sli_performance = round((within_target * 100.0) / executions,
decimals: 2)
| fieldsAdd slo_target = 95.0 // 95% of requests < 3s
| fieldsAdd slo_status = if(sli_performance >= slo_target, "✅ Met", else:
"❌ Breached")
| sort sli_performance asc
| limit 30
```

```
// Error budget burn rate (last 7 days)
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd day = formatTimestamp(timestamp, format: "yyyy-MM-dd")
| summarize {
    total = count(),
    failed = countIf(synthetic.availability == false)
}, by: {dt.entity.synthetic_test, day}
| fieldsAdd daily_error_rate = round((failed * 100.0) / total, decimals: 3)
| fieldsAdd budget_consumed = round(failed * 100.0 / (total * 0.001),
decimals: 1) // vs 0.1% budget
| sort dt.entity.synthetic_test, day desc
```

6. Alerting Strategies

Alert Types

Alert Type	Trigger	Use Case
Availability	Consecutive failures	Outage detection
Performance	Response time threshold	Degradation
SLO	Error budget consumption	Proactive warning
SSL	Certificate expiration	Security

Alert Configuration Best Practices

Setting	Recommendation	Reason
Consecutive failures	2-3	Avoid false positives
Location threshold	2+ locations	Confirm not local issue
Alert delay	5-10 minutes	Allow for transients
Auto-resolve	After 2 successes	Clear resolved issues

```
// Monitors with recent failures (potential outages)
// Check for monitors that have failed multiple times recently
fetch bizevents, from: now() - 1h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| summarize {
    failure_count = count(),
    last_failure = max(timestamp),
    first_failure = min(timestamp)
}, by: {dt.entity.synthetic_test, dt.entity.synthetic_location}
| filter failure_count >= 2
| sort failure_count desc
| limit 20
```

```
// Performance degradation alerts (P95 > threshold)
fetch bizevents, from: now() - 1h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    p95_response_ms = percentile(toDouble(synthetic.response_time), 95),
    executions = count()
}, by: {dt.entity.synthetic_test}
| filter p95_response_ms > 5000 // > 5 seconds threshold
| sort p95_response_ms desc
```

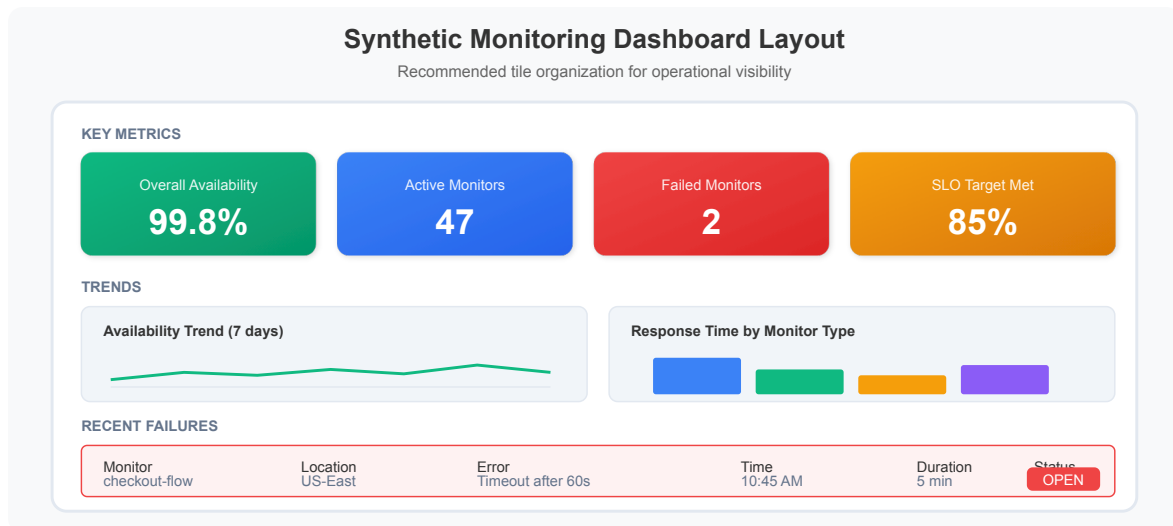
```
// Multi-location failure detection (global outage indicator)
fetch bizevents, from: now() - 30m
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| summarize {
    failing_locations = countDistinct(dt.entity.synthetic_location),
    failure_count = count(),
    locations = collectDistinct(dt.entity.synthetic_location)
}, by: {dt.entity.synthetic_test}
| filter failing_locations >= 2
| fieldsAdd severity = if(failing_locations >= 3, "CRITICAL", else:
"WARNING")
| sort failing_locations desc
```

7. Dashboard Building

Recommended Dashboard Tiles

Tile Type	Content	Visualization
Overall Health	Availability %	Single value
Availability Trend	Hourly availability	Line chart
Response Time	P95 by monitor	Bar chart
Location Map	Geographic availability	World map
Failures Table	Recent failures	Table
SLO Status	Error budget	Gauge

Dashboard Layout



```
// Dashboard tile: Overall availability (single value)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| summarize {
    availability_pct = round(countIf(synthetic.availability == true) * 100.0
/ count(), decimals: 2)
}
```

```
// Dashboard tile: Active vs failed monitors
fetch bizevents, from: now() - 1h
| filter event.provider == "dynatrace.synthetic"
| summarize {
    latest_status = takeFirst(synthetic.availability)
}, by: {dt.entity.synthetic_test}
| summarize {
    total_monitors = count(),
    healthy = countIf(latest_status == true),
    failing = countIf(latest_status == false)
}
```

```
// Dashboard tile: Response time by monitor (bar chart)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    p95_response_ms = percentile(toDouble(synthetic.response_time), 95)
}, by: {dt.entity.synthetic_test}
| sort p95_response_ms desc
| limit 10
```

```
// Dashboard tile: Recent failures table
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| fields timestamp,
    monitor = dt.entity.synthetic_test,
    location = dt.entity.synthetic_location,
    error = synthetic.error_message
| sort timestamp desc
| limit 20
```

```
// Dashboard tile: Hourly availability trend (line chart)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd hour_bucket = bin(timestamp, 1h)
| summarize {
    success_count = countIf(synthetic.availability == true),
    total_executions = count(),
    failures = countIf(synthetic.availability == false)
}, by: {hour_bucket}
| fieldsAdd availability_pct = round((success_count * 100.0) /
total_executions, decimals: 2)
| sort hour_bucket asc
```

Summary

In this notebook, you learned:

- ✅ **Availability analysis** - Calculating and trending availability
- ✅ **Performance analysis** - Percentiles, timing breakdown, anomalies
- ✅ **Location comparison** - Geographic performance analysis

- ✓ **SLO configuration** - Service level objectives and error budgets
 - ✓ **Alerting strategies** - Outage detection, performance alerts
 - ✓ **Dashboard building** - Key tiles and layouts
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Series Complete!

You have completed the Synthetic Monitoring Best Practices series:

1. ✓ **SYNTH-01:** Fundamentals
 2. ✓ **SYNTH-02:** Browser Monitors
 3. ✓ **SYNTH-03:** HTTP Monitors
 4. ✓ **SYNTH-04:** Private Locations
 5. ✓ **SYNTH-05:** Network Monitoring
 6. ✓ **SYNTH-06:** Analytics & Alerting
-

References

- [Synthetic Monitoring Overview](#)
 - [Analyze Synthetic Monitors](#)
 - [Service Level Objectives](#)
 - [Alerting Profiles](#)
 - [Dashboards](#)
-

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